Whole-brain TRAP —candidate stories

A) Dual-role craving (Active), B) Passive withdrawal (patient-relevant)

Two findings — why keep both

- FINDING A — ORBm & BMAp: dual-mode craving regions (Active >> Passive).
- Opioid seeking at Post + negative-affect craving persists in Withdrawal. Passive did not show any change

- FINDING B — COAa: Passive > Active at Post and Withdrawal (patient-relevant). Best anatomical fit in cluster 3 for cue/context + withdrawal-state → generalized motivation candidate

Why finding A and finding B targets?

To solve the problem: Opioid seeking vs general seeking (To exclude general seeking possibility)

Why ORBm/BMAp?

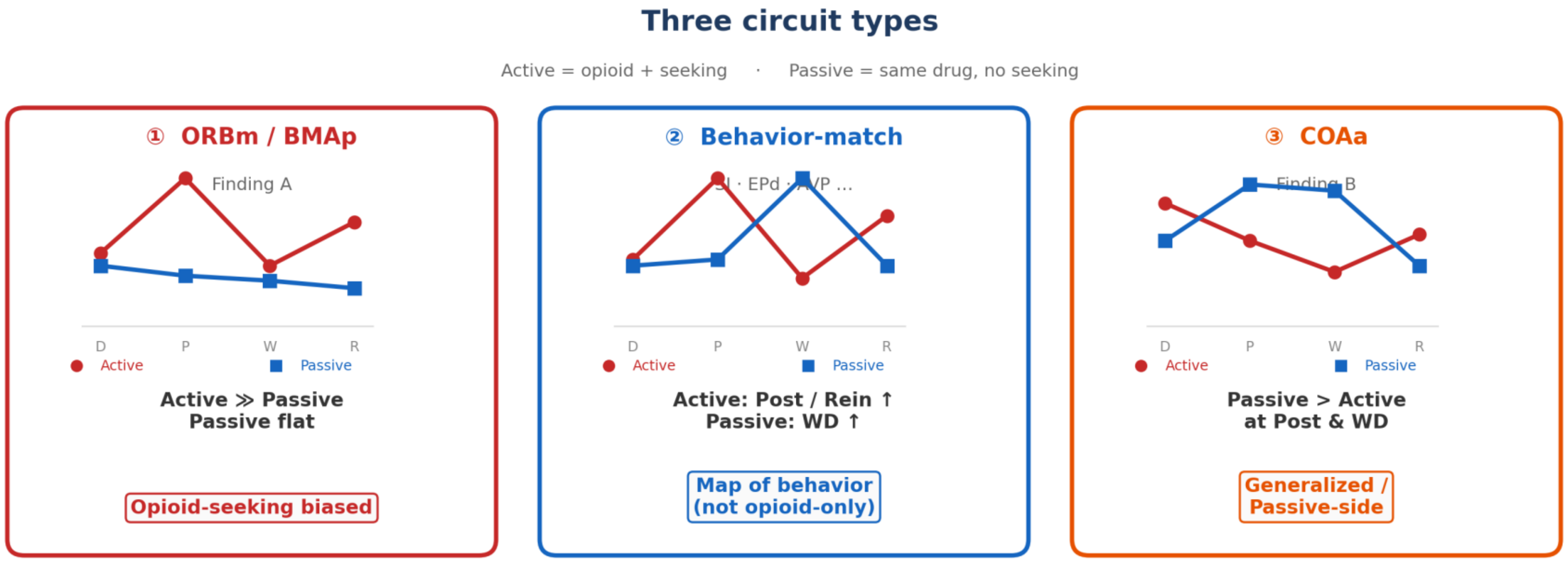
*They stay **Active ≫ Passive** with Passive flat. Shared general-seeking / context-only accounts predict similar Passive trajectories; we don't see that. So these regions **favor opioid-related Active seeking** and **partly exclude** pure general-seeking circuits.*

Why not behavior-matched regions for Finding A?

*Regions that match behavior patter (Active Post/Rein high; Passive WD high, often > Active) change in **both** delivery modes. That means they track **opioid craving and Passive generalized motivation** — useful as a map, but not opioid-selective.*

Why COAa?

*Passive > Active at Post and Withdrawal — the epochs of context + withdrawal state in our schematic. COAa is a **generalized / Passive-side** candidate: manipulating it may affect **broad seeking** (e.g. sucrose), whereas ORBm/BMAp should spare that more. WD is not pure sucrose, so this remains a **testable prediction**, not a proven dissociation.*



Analysis pipeline Roadmap — why cluster first, then zoom in

Funnel logic: hundreds of regions → 4 clusters → 2 behaviour-relevant clusters → 7 regions → zoom-in-> only 2 brain regions : ORBm, BMAp

Problem: the brain has hundreds of regions. Testing each one individually is under-powered and hard to interpret (multiple-comparisons, no structure).

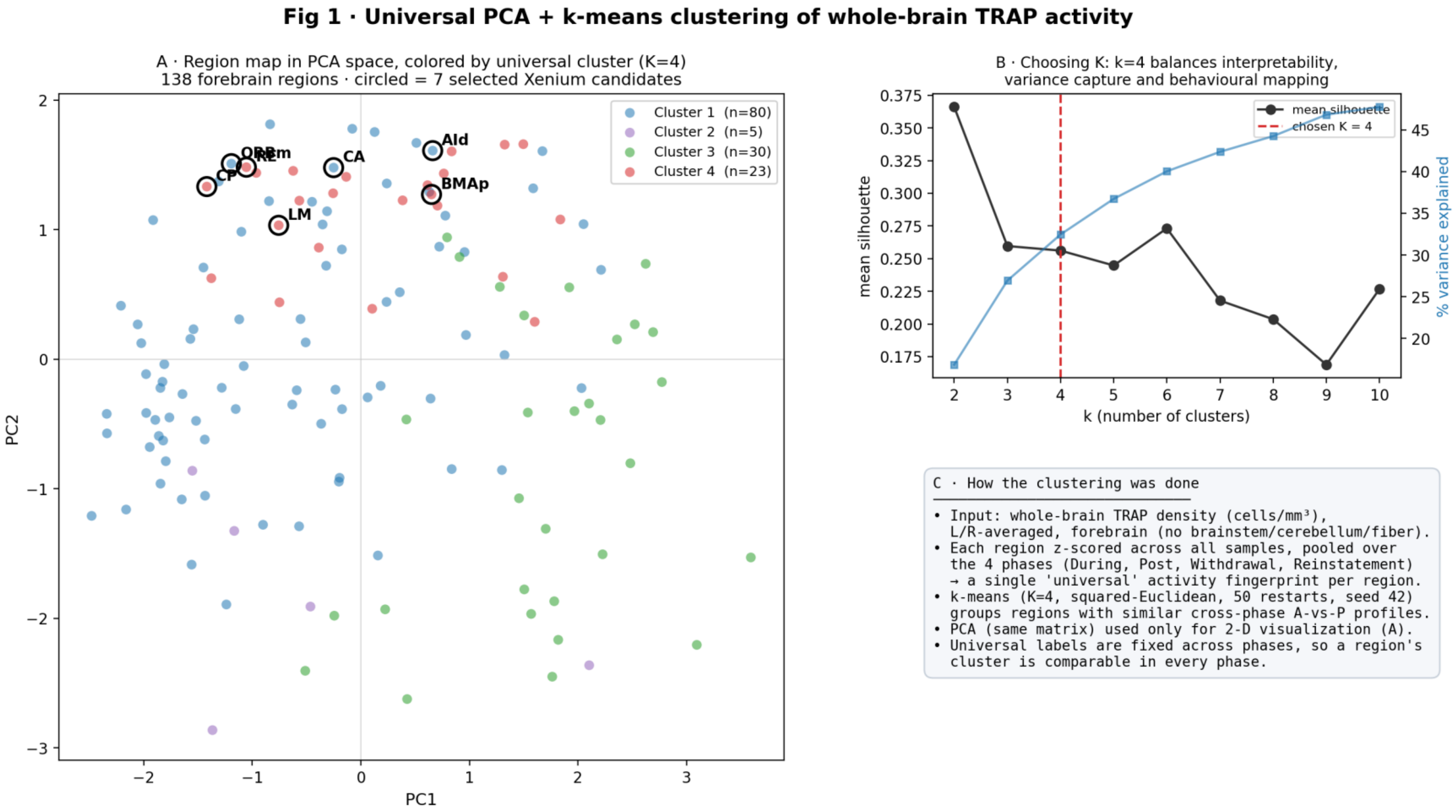
Step 1 — FILTER by clustering: pool all forebrain regions (L/R-averaged) and group them by their Active-vs-Passive activity fingerprint across all 4 phases (universal PCA + k-means, K=4). Hundreds of regions → 4 functional clusters.

Step 2 — KEEP behaviourally-aligned clusters: only clusters 1 & 4 reproduce the motivation timeline (Active > Passive at Post & Reinstatement). → shortlist of candidate regions.

Step 3 — SELECT **active always higher than passive regions**: BMAp, LM, RE, CP (cluster 4) + ORBm, CA, AId (cluster 1) = 7 regions, agreeing with an independent top-N ranking.

Step 4 — ZOOM IN: for each of the 7 regions, look at density AND cell count across phases, Active vs Passive, showing every individual mouse. **After ZOOM IN only ORBm and BMAp survived** while others are less promising

Fig 1 · How clustering was done + what clusters look like



Method (Fig 1)

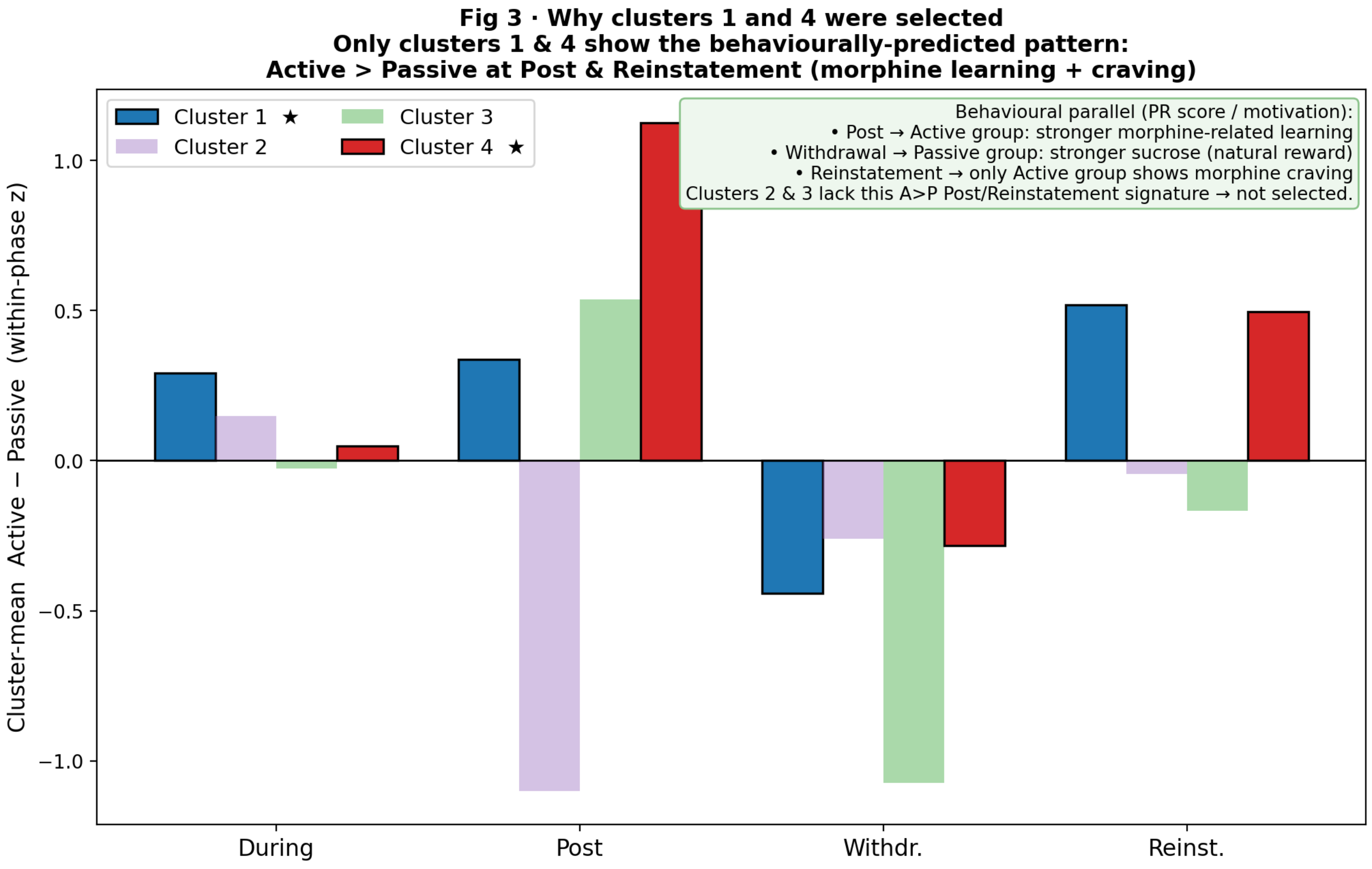
A: 138 forebrain regions in PCA space, colored by cluster (K=4); circled = 7 selected.

B: k was chosen at 4 (balances interpretability, variance, behaviour mapping).

C: each region z-scored across samples (4 phases pooled) → k-means (sq-Euclidean, 50 restarts).

PCA is only for 2-D display; clusters use the full profile.

Fig 3 · Why clusters 1 and 4 were selected



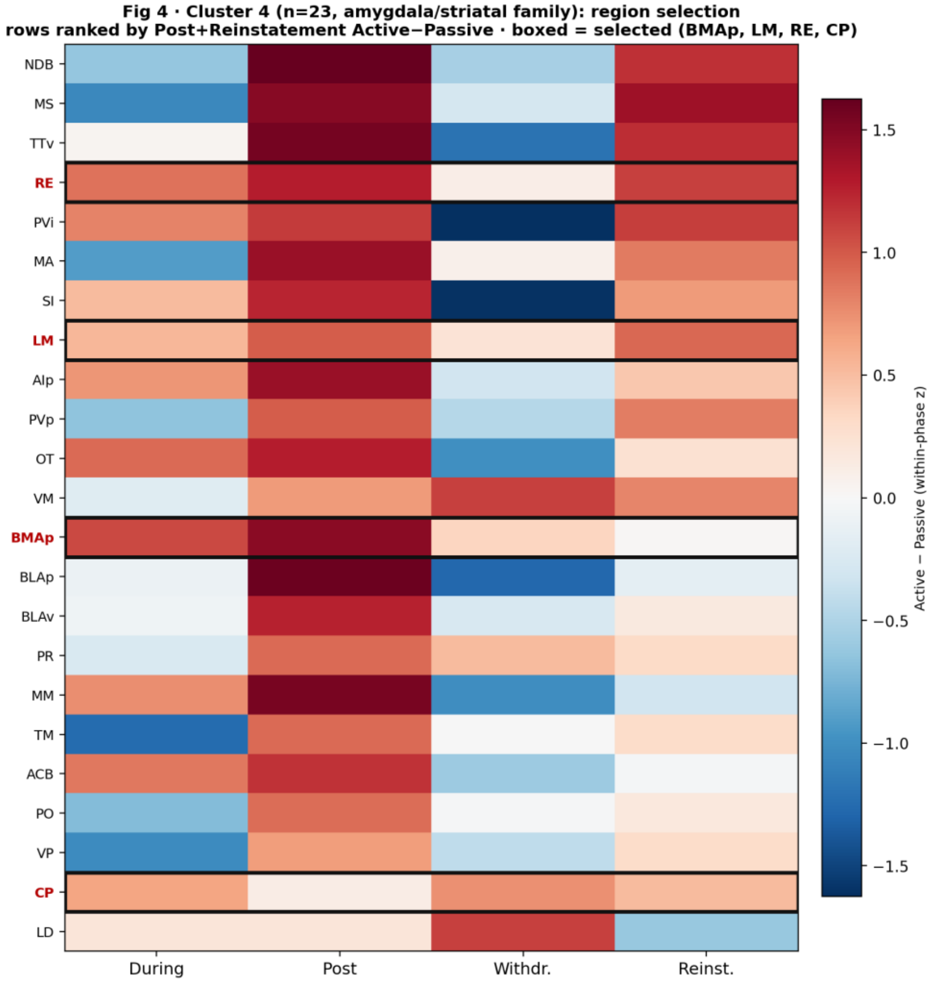
Selection of clusters (Fig 3)

Per-phase bars of cluster-mean Active – Passive.

Only clusters 1 & 4 show the predicted pattern (A>P at Post & Reinstatement).

Post = Active morphine learning; Withdrawal = Passive sucrose; Reinstatement = Active craving.

Fig 4 · Cluster 4 — why BMAp, LM, RE, CP



Selection rule (3 criteria, not score alone): (1) member of behaviourally-aligned cluster 4; (2) Active > Passive in the craving-relevant phases (Post/Reinstatement); (3) canonical addiction-circuit node with a defined manipulation target. Other statistically-elevated cluster-4 regions (e.g. NDB, MS, TTv) were deprioritised as less specific / harder to target. BMAp, LM, RE, CP satisfy all three and agree with the independent top-N ranking.

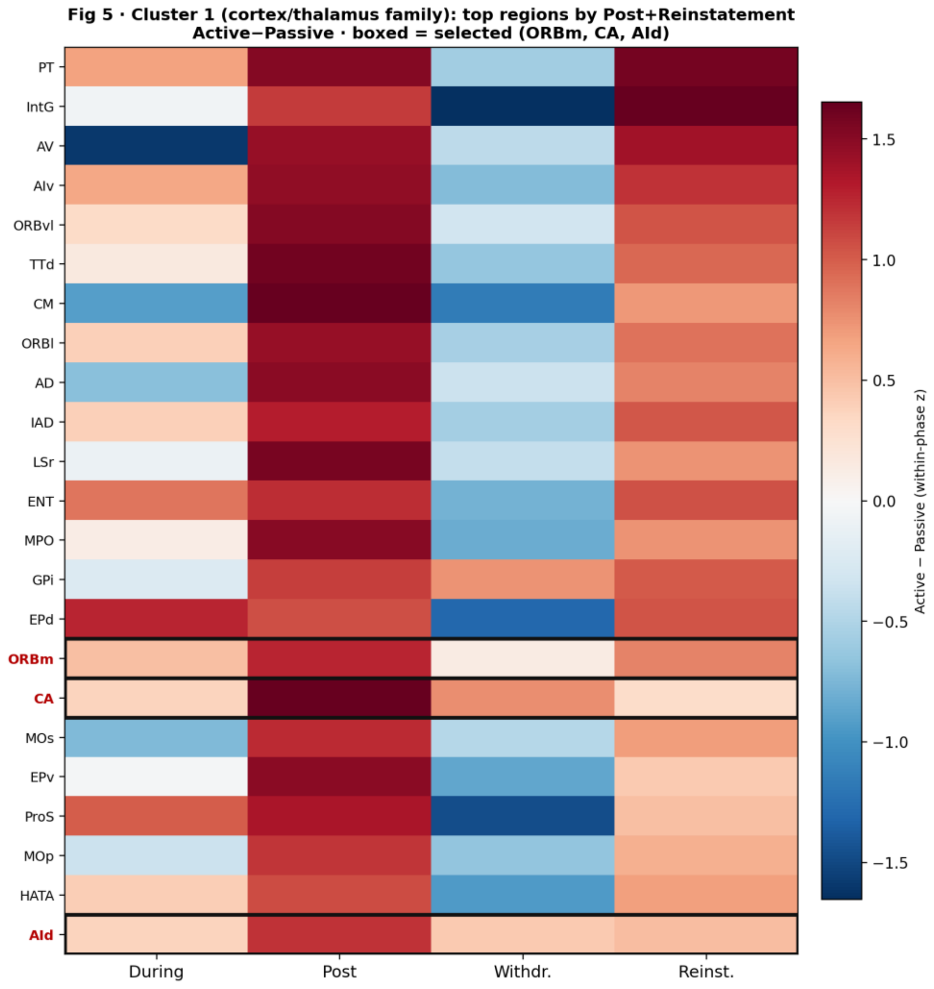
Within cluster 4 (Fig 4)

cluster-4 regions × phase heatmap of Active − Passive; boxed = selected.

Selection = A>P in craving phases

- FINDING A -mixed craving regions (Active >> Passive).

Fig 5 · Cluster 1 — why ORBm, CA, AId



Selection rule (3 criteria, not score alone): (1) member of behaviourally-aligned cluster 1; (2) Active > Passive at Post/Reinstatement; (3) canonical addiction-relevant cortical/limbic node. Higher-scoring but less interpretable nodes (e.g. PT, IntG, AV thalamic nuclei) were deprioritised. ORBm, CA, AId satisfy all three and agree with the independent top-N ranking.

Within cluster 1 (Fig 5)

Top cluster-1 regions by Post+Reinstatement Active − Passive; boxed = selected.

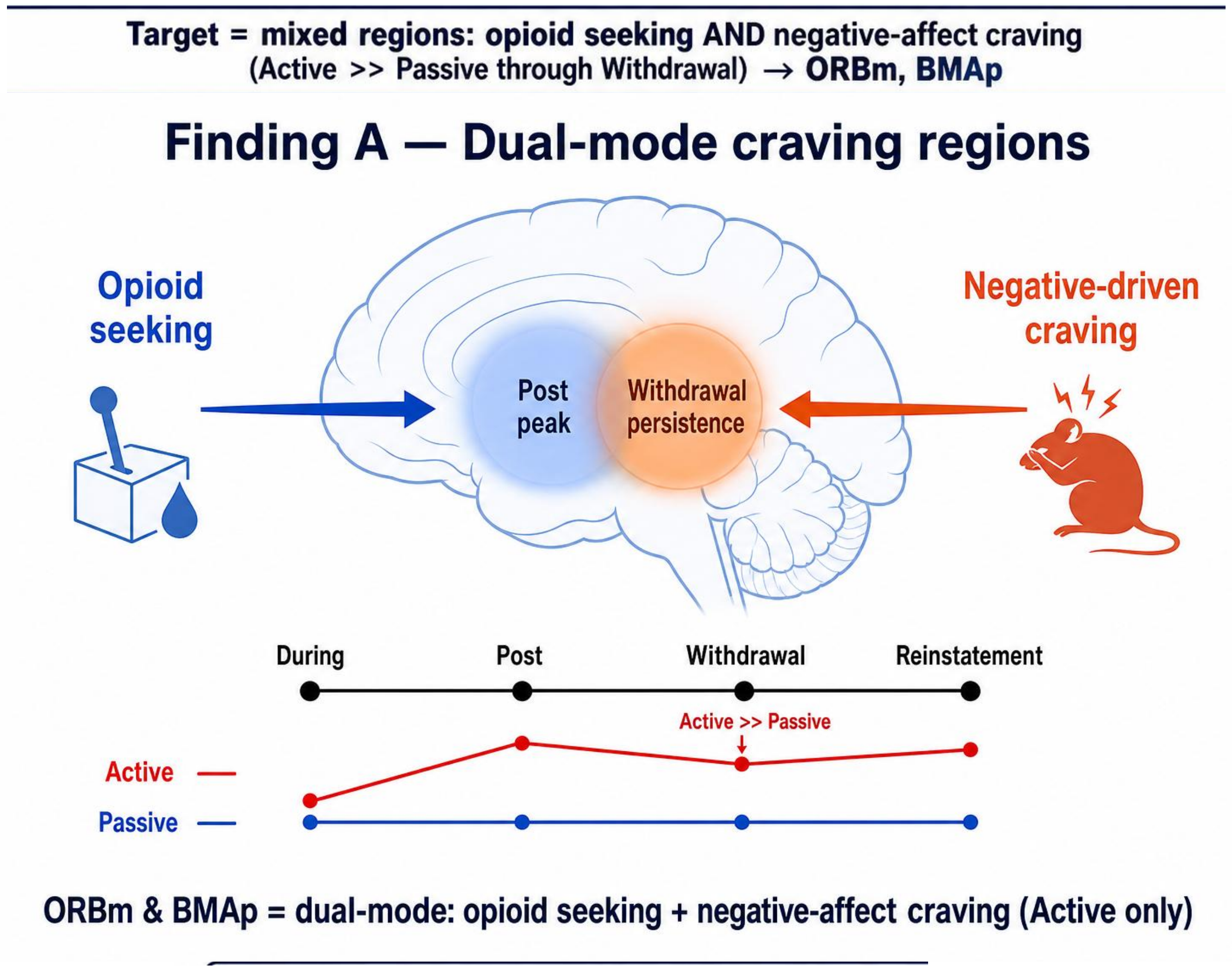
- FINDING A -mixed craving regions (Active >> Passive).

Part 2 · Zoom-in on each selected region (group level)

density + cell count · Active vs Passive · individual mice shown

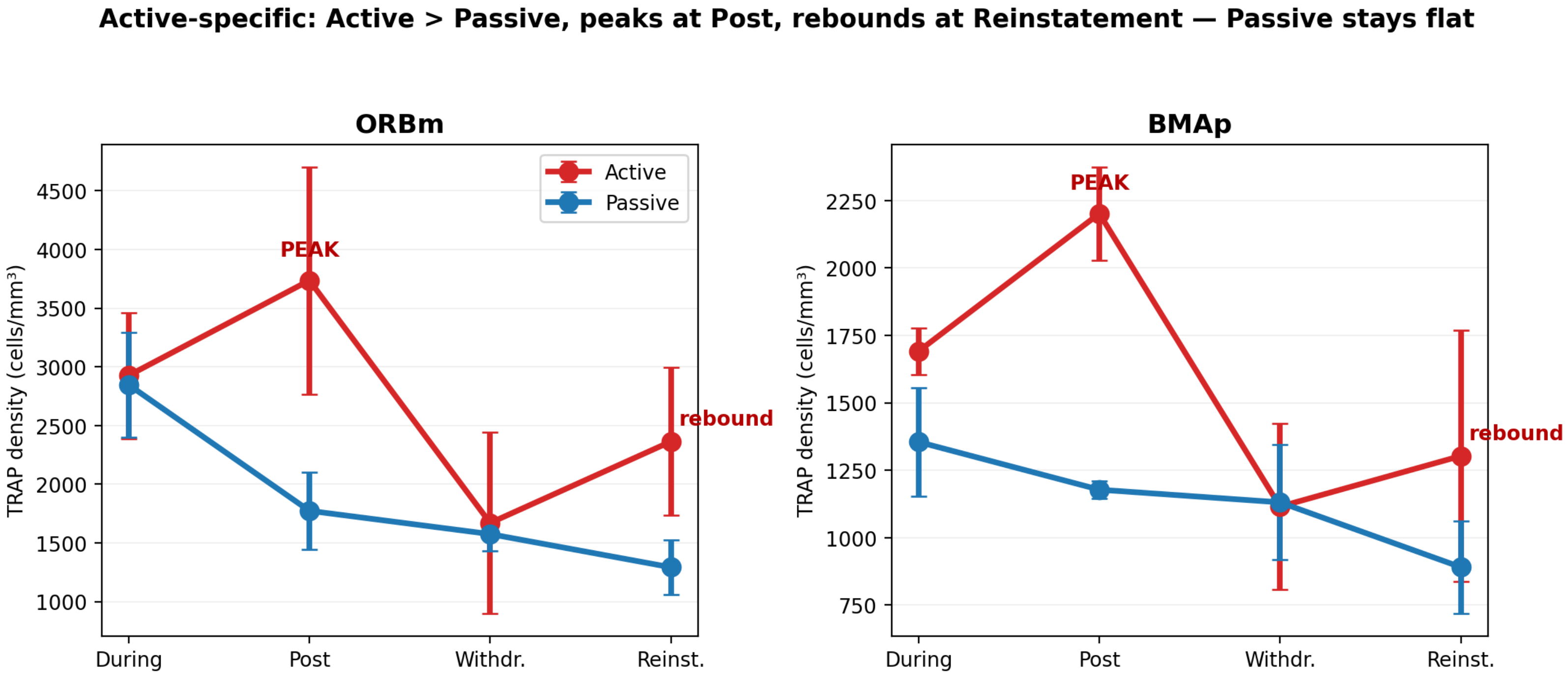
For each of the 7 regions I plot BOTH density (cells/mm^3) and cell count across the 4 phases, separately for Active and Passive.

Finding A — Why Active >> Passive even in Withdrawal?



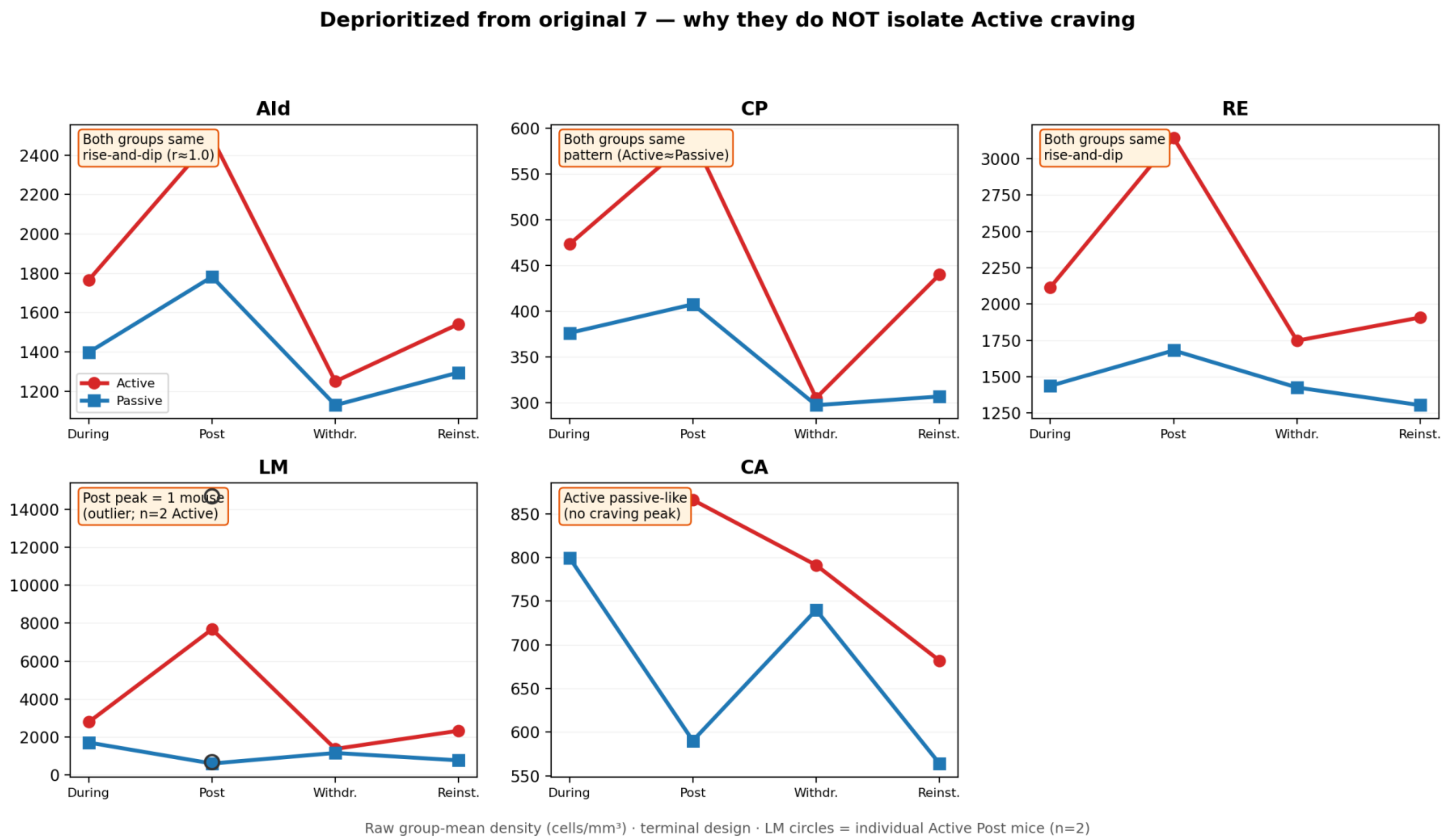
- Not just Post craving — Withdrawal matters for incubation.
- Negative affect during Withdrawal can drive craving increase.
- Target = mixed regions: opioid seeking + negative-driven craving.
- Active >> Passive through Withdrawal (Passive flat).
- → ORBm, BMAp = dual-role craving regions.

Finding A — ORBm & BMAp (raw density)



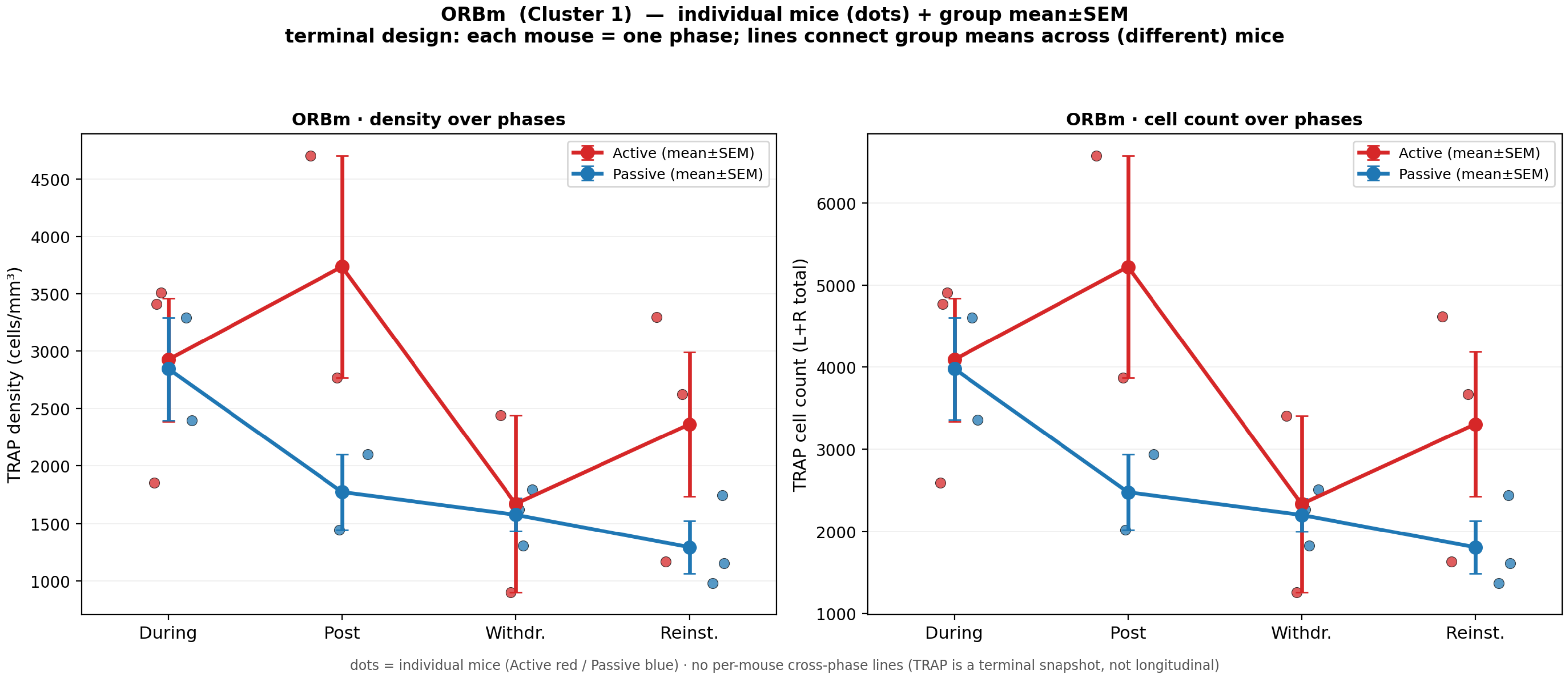
- Active PEAKS at Post.
- Dips at Withdrawal but stays >> Passive.
- REBOUNDS at Reinstatement.
- Passive flat — no craving pattern.

Why not the other original-7 regions?

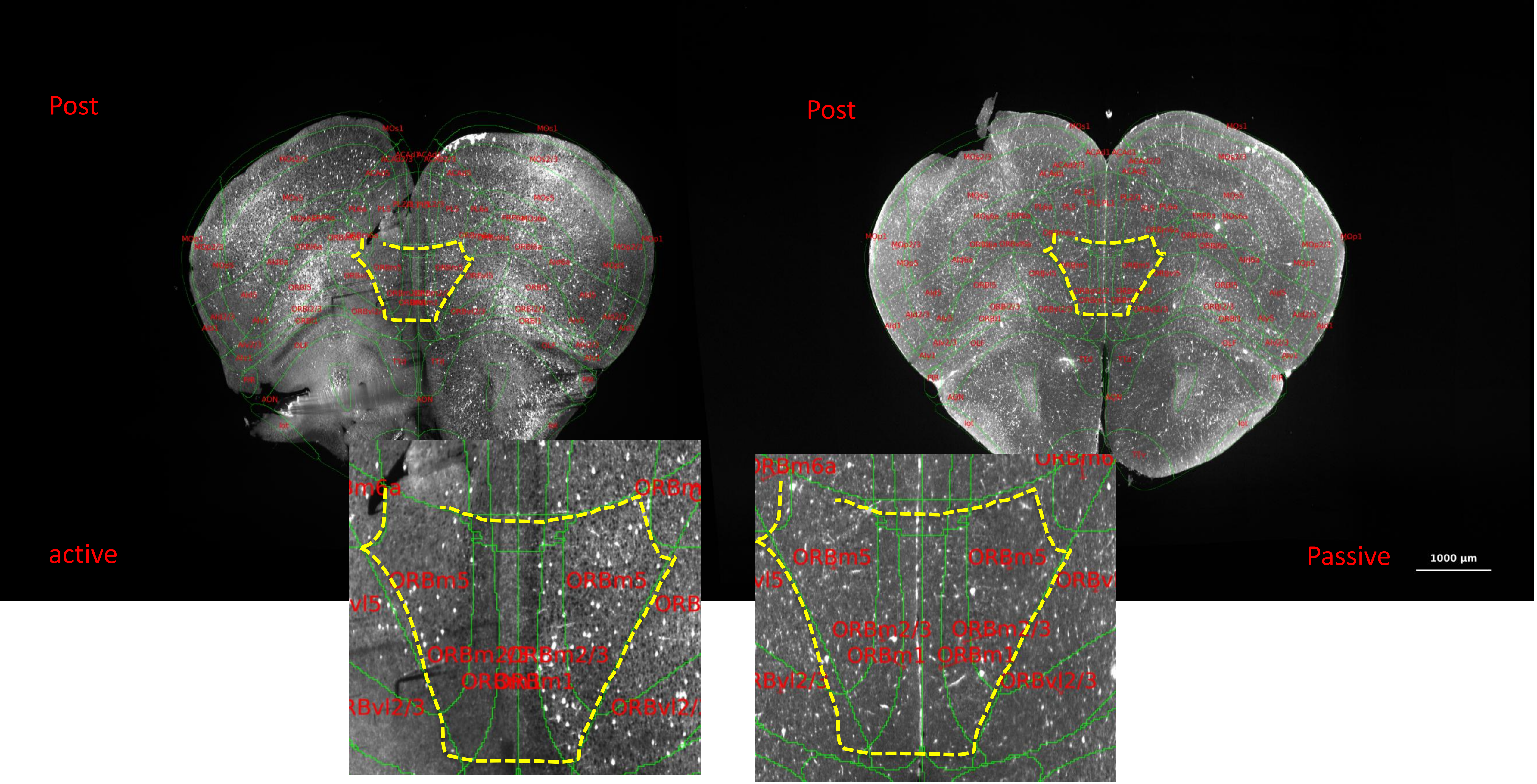


- AId, CP, RE: Active ≈ Passive — same rise-and-dip (cannot isolate craving).
- AId = weakest: trajectories nearly identical (r ≈ 1.0).
- CA: Active is passive-like — no clean Post craving peak.
- LM: Post "peak" driven by 1 mouse (n=2 Active at Post).
- → ORBm & BMAp kept as clean Active-only dual-mode regions.

Finding A — ORBm zoom-in (density & cell count)

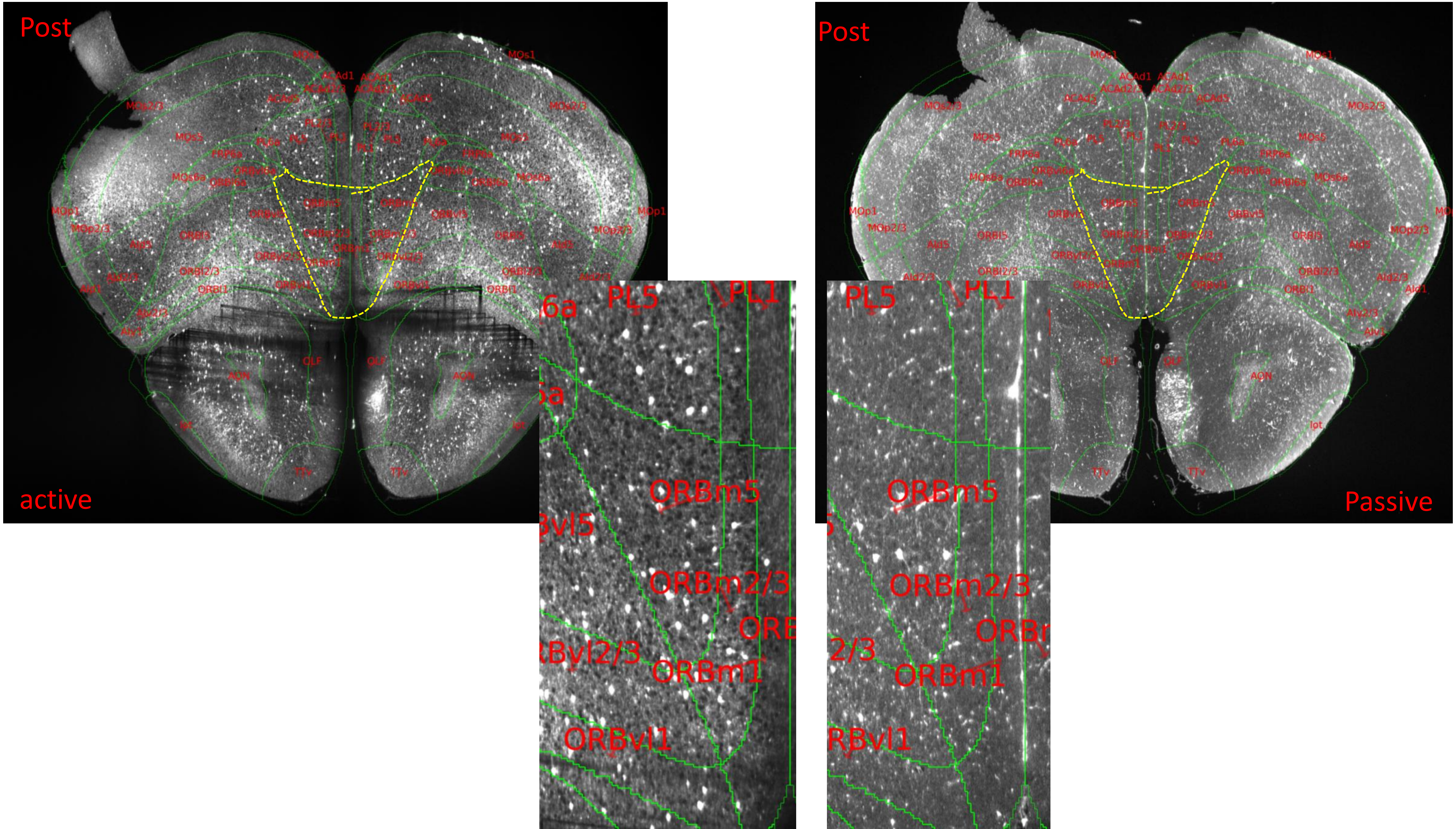


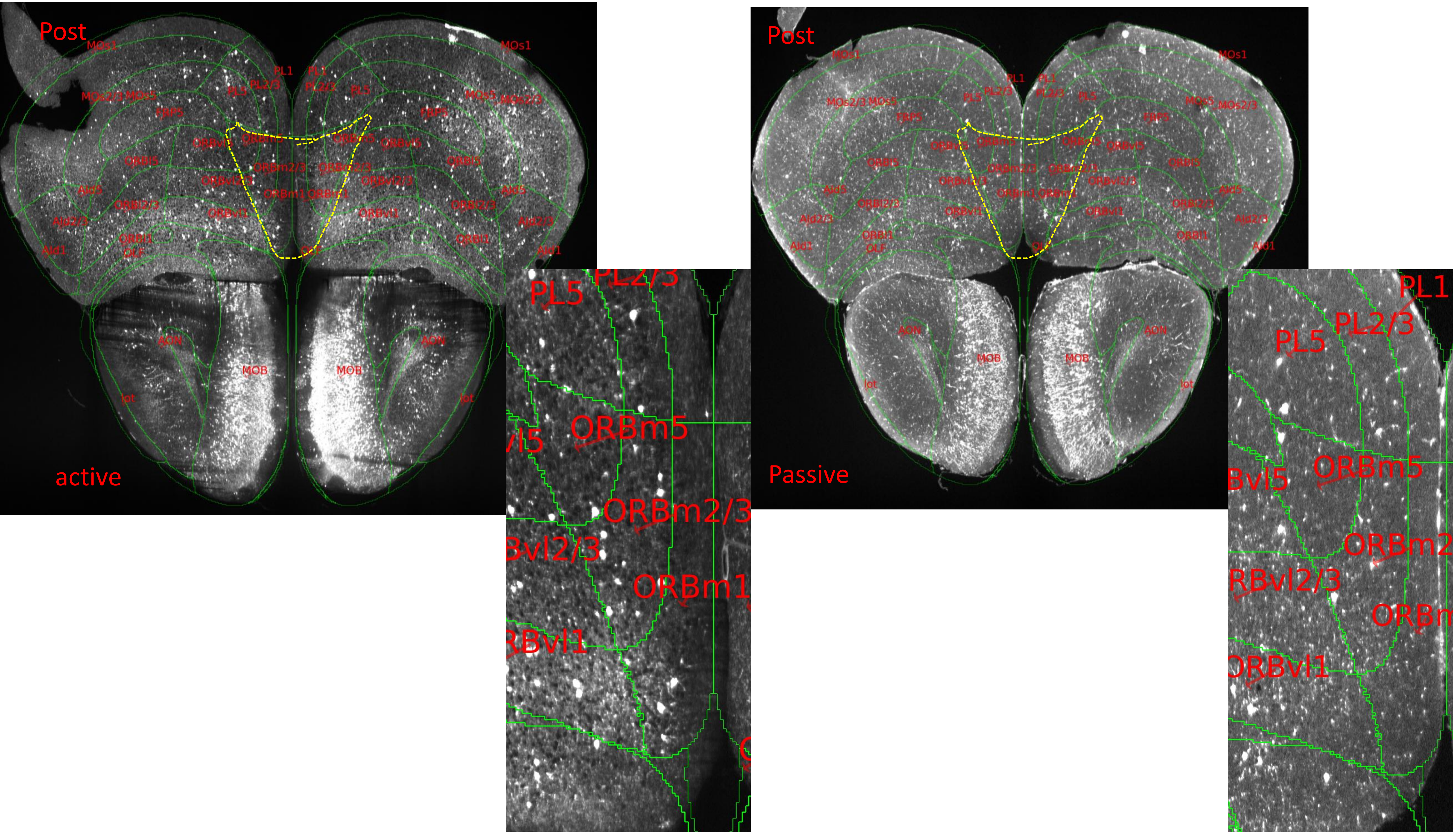
Group trajectories. Active peaks Post, dips Withdrawal, rebounds; Passive flat.


Post
Post
active
Passive
1000 µm
ORBm5
ORBm5
ORBm2/3
ORBm1
ORBvl2/3
ORBm6a
ORBm5
ORBm5
ORBm2/3
ORBm2/3
ORBm1
ORBm1
ORBvl2/3

ORBm "Post" phase

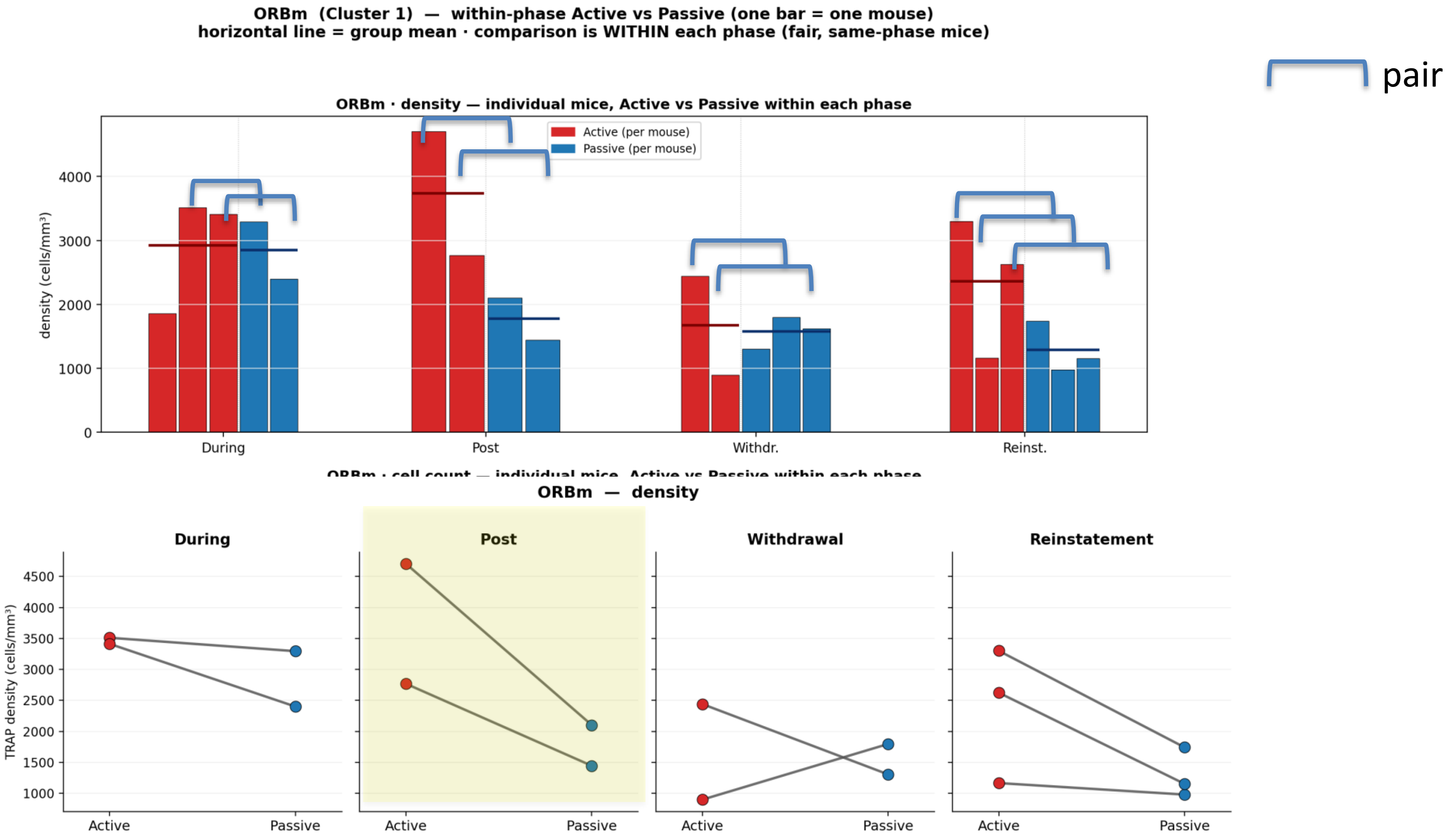
Active mouse (left) vs Passive (right)





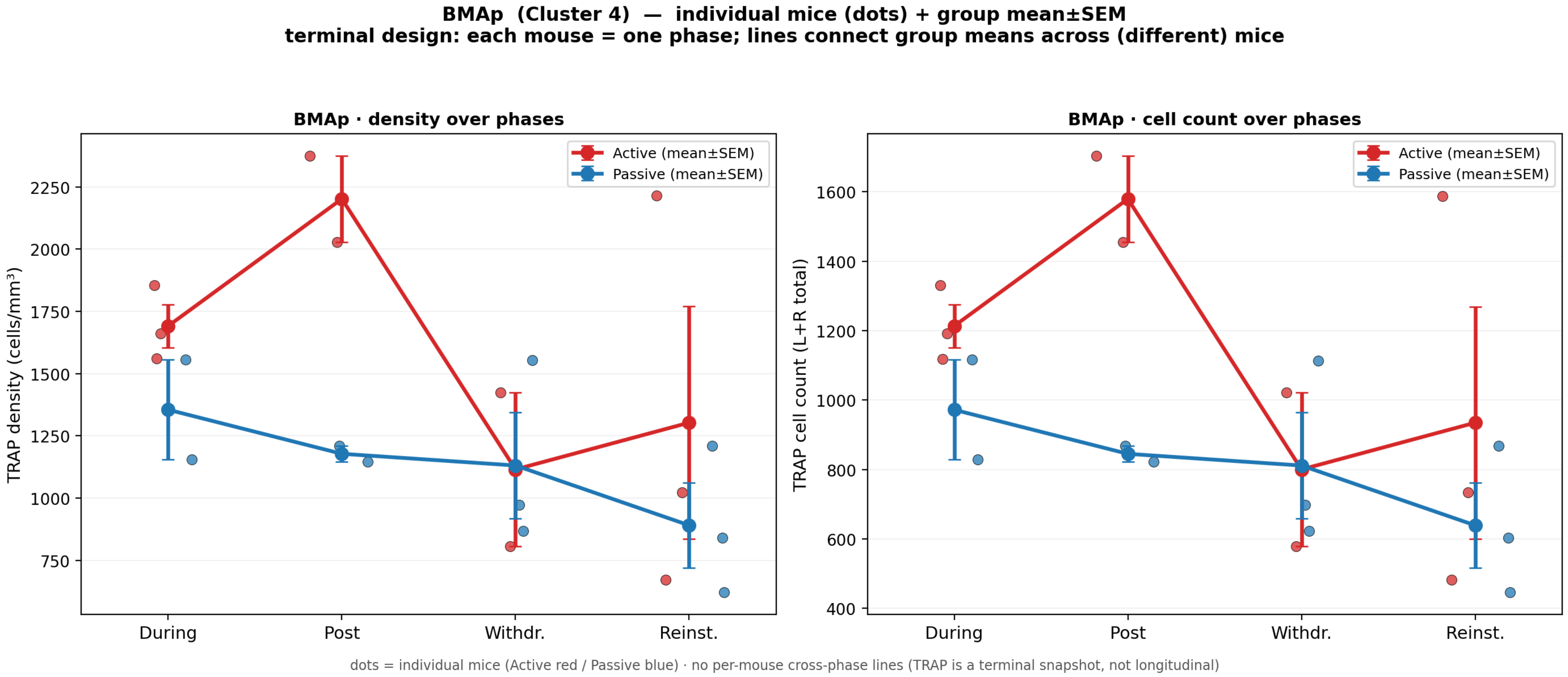
Post
active
Post
Passive
PL5
PL2/3
ORBm5
ORBm2/3
ORBm1
PL1

Finding A — ORBm within-phase (one bar = one mouse)

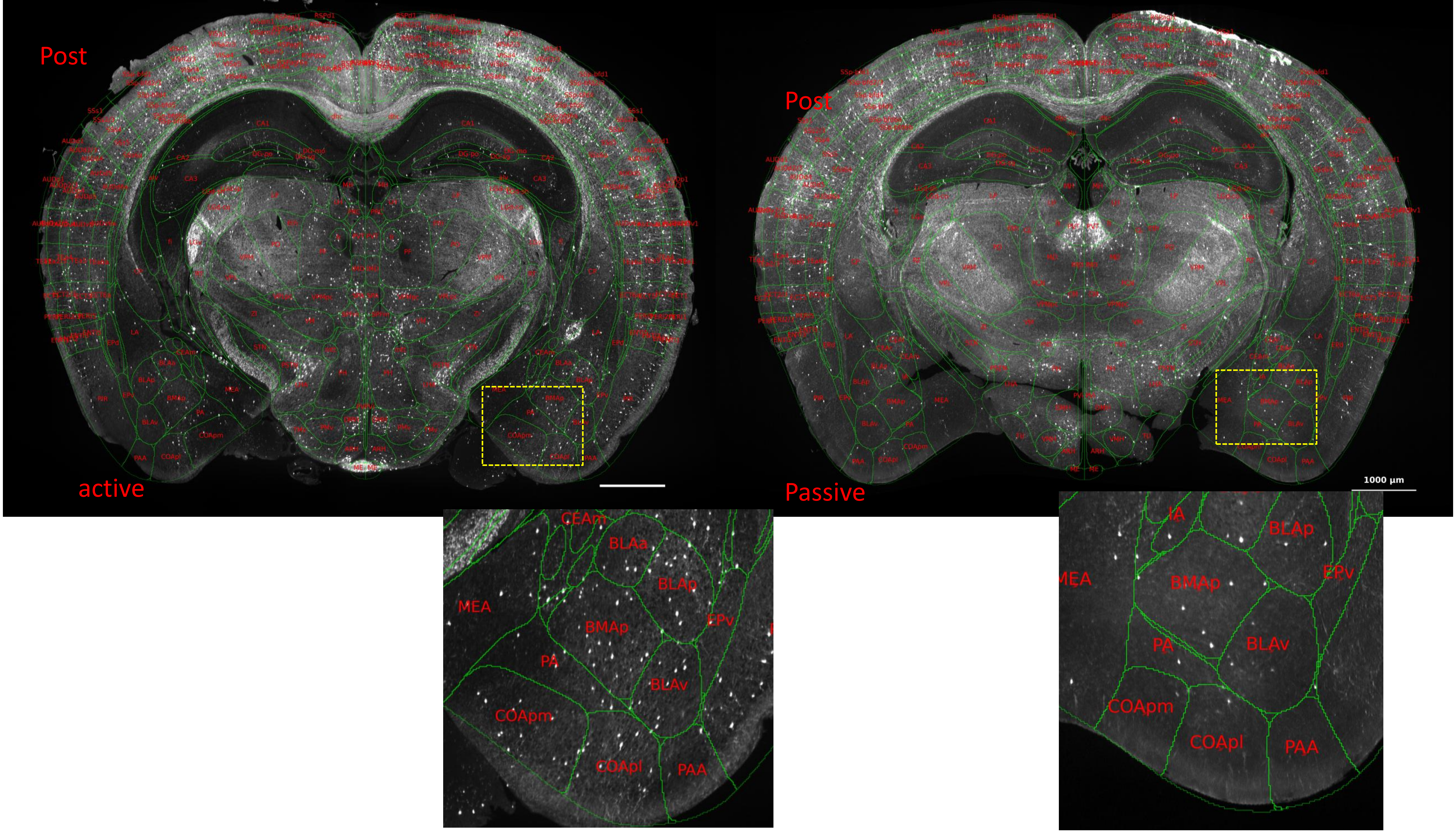


Within each phase: every Active mouse vs every Passive mouse. Red = Active, blue = Passive.

Finding A — BMAp zoom-in (density & cell count)

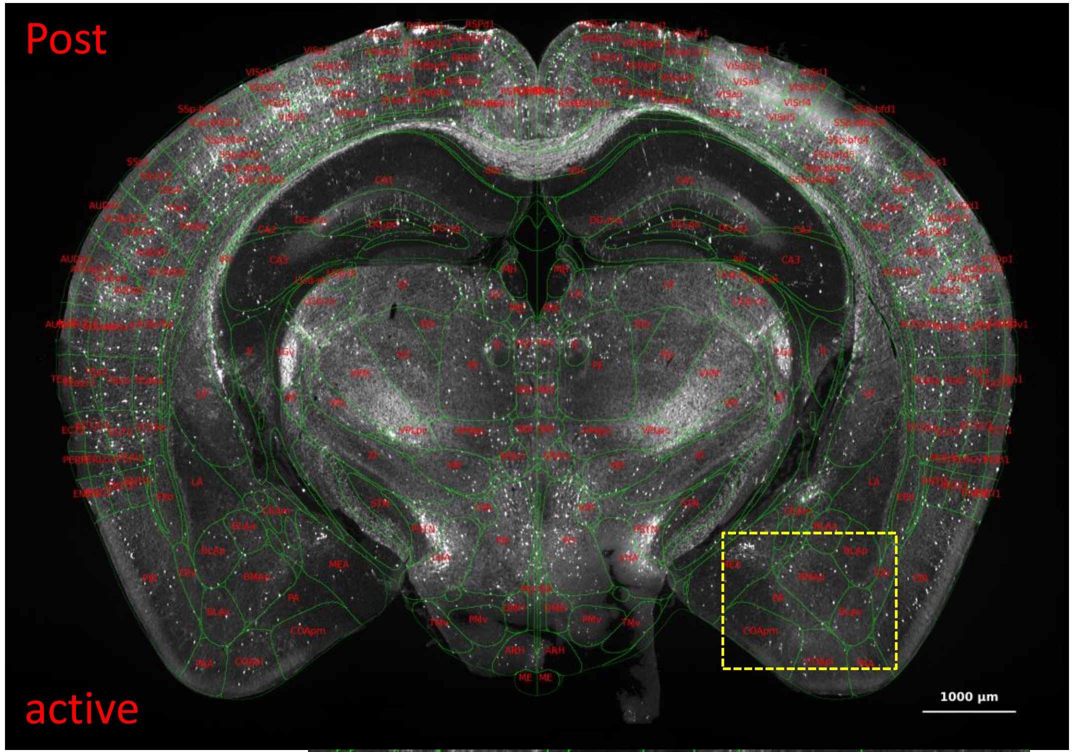


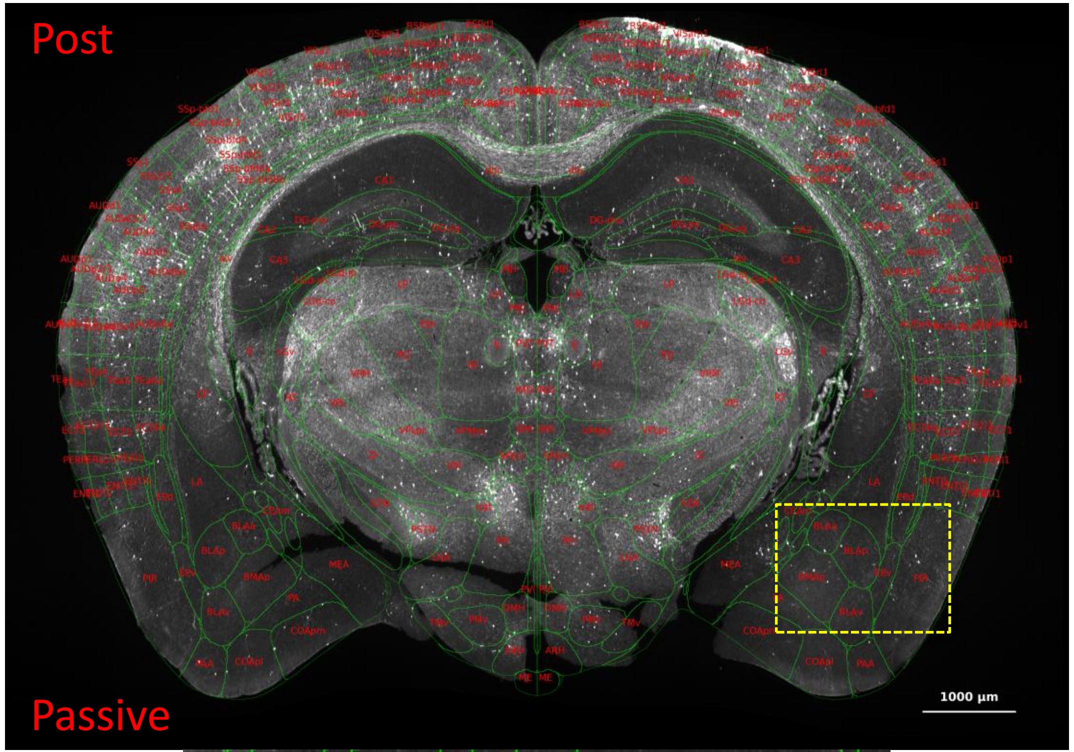
Group trajectories. Active peaks Post, dips Withdrawal, rebounds; Passive flat.

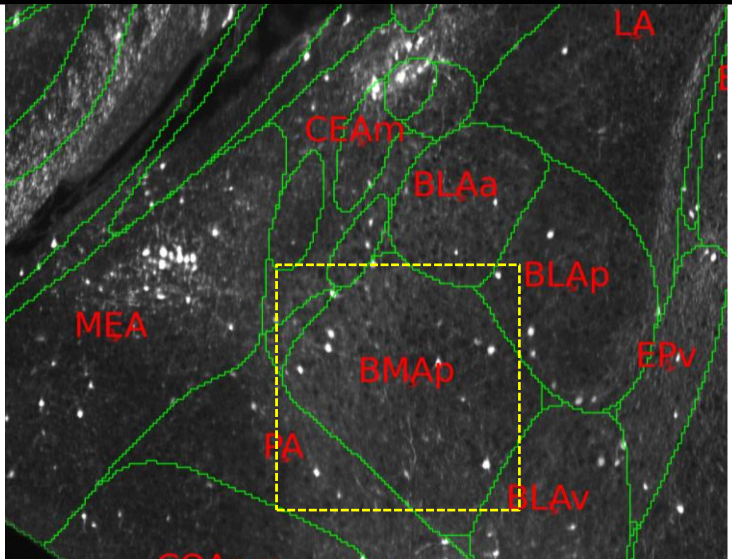

Post
active
Post
Passive
1000 µm
CEAm
BLAa
BLAp
MEA
BMAp
EPv
PA
BLAv
COApm
COApl
PAA
IA
BLAp
MEA
BMAp
EPv
PA
BLAv
COApm
COApl
PAA

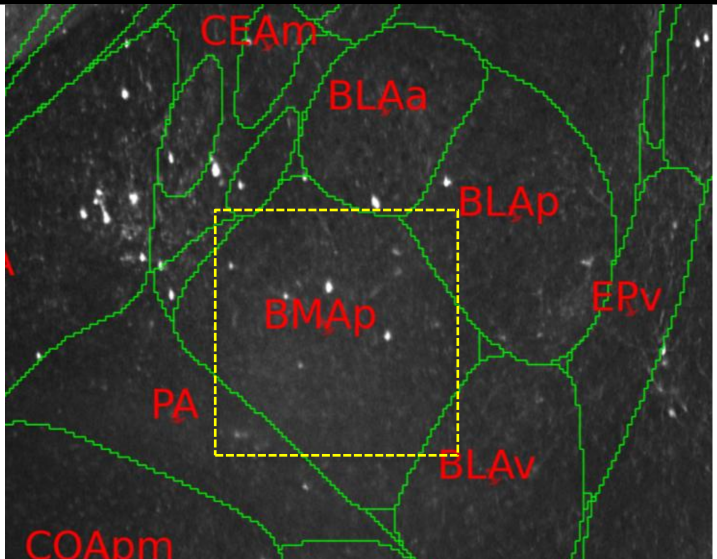
BMAp “Post” phase

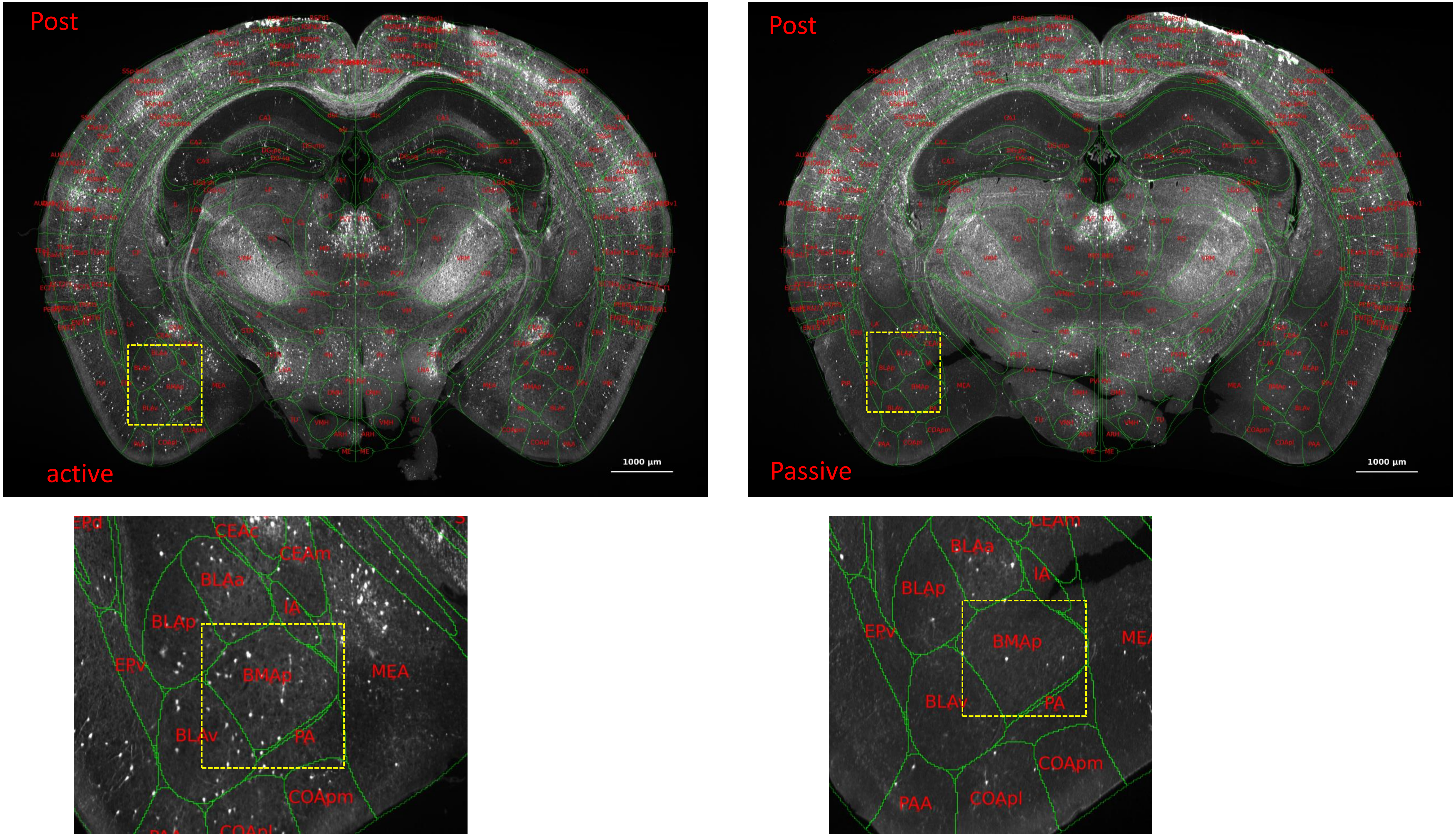
Active mouse vs Passive mouse





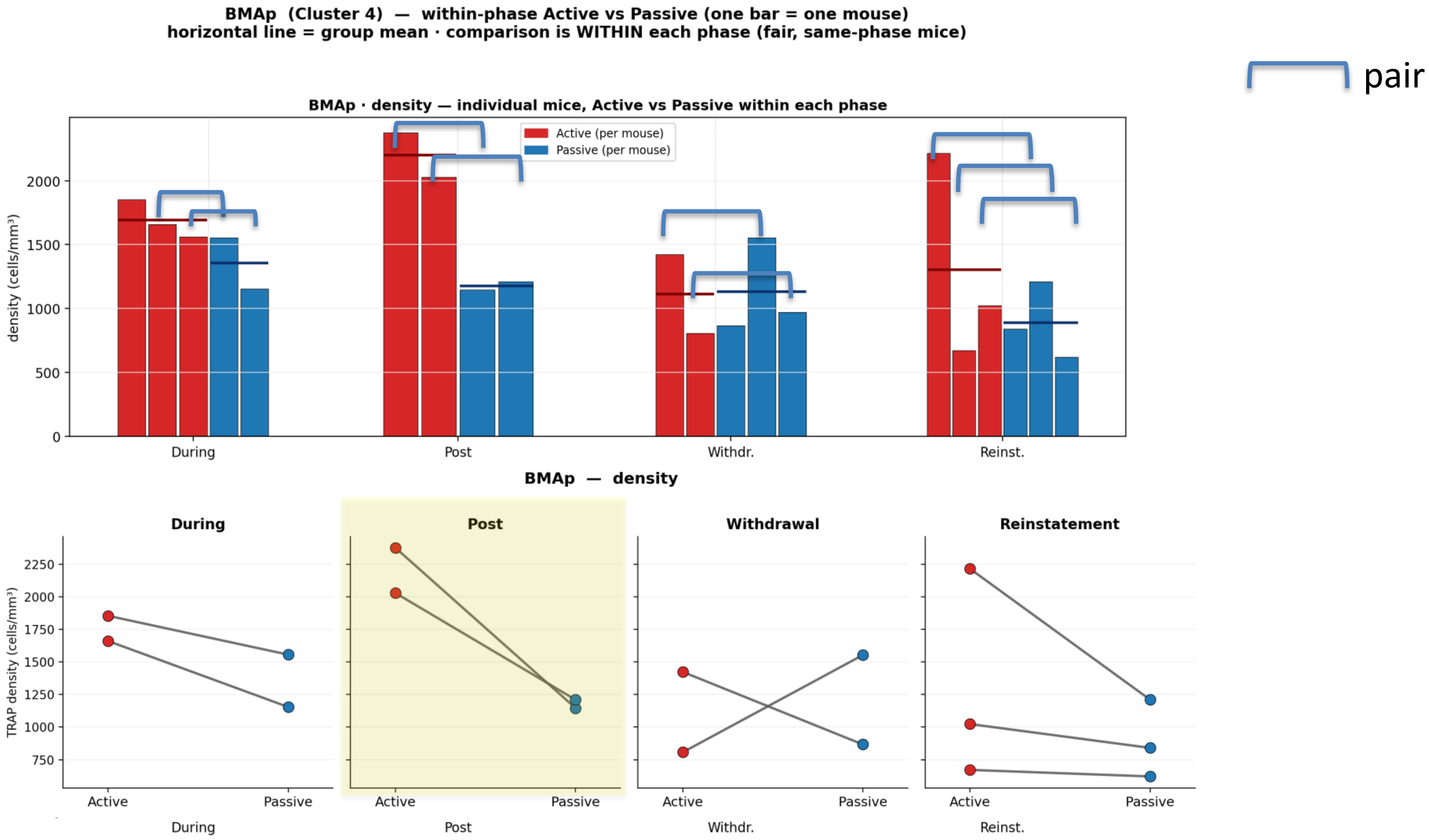






Post
active
1000 µm
Post
Passive
1000 µm
EPd
CEAc
CEAm
BLAa
IA
BLAp
EPv
BMAp
MEA
BLAv
PA
COApm
COApl
CEAm
BLAa
IA
BLAp
EPv
BMAp
MEA
BLAv
PA
COApm
PAA
COApl

Finding A — BMAp within-phase (one bar = one mouse)



Within each phase: every Active mouse vs every Passive mouse. Red = Active, blue = Passive.

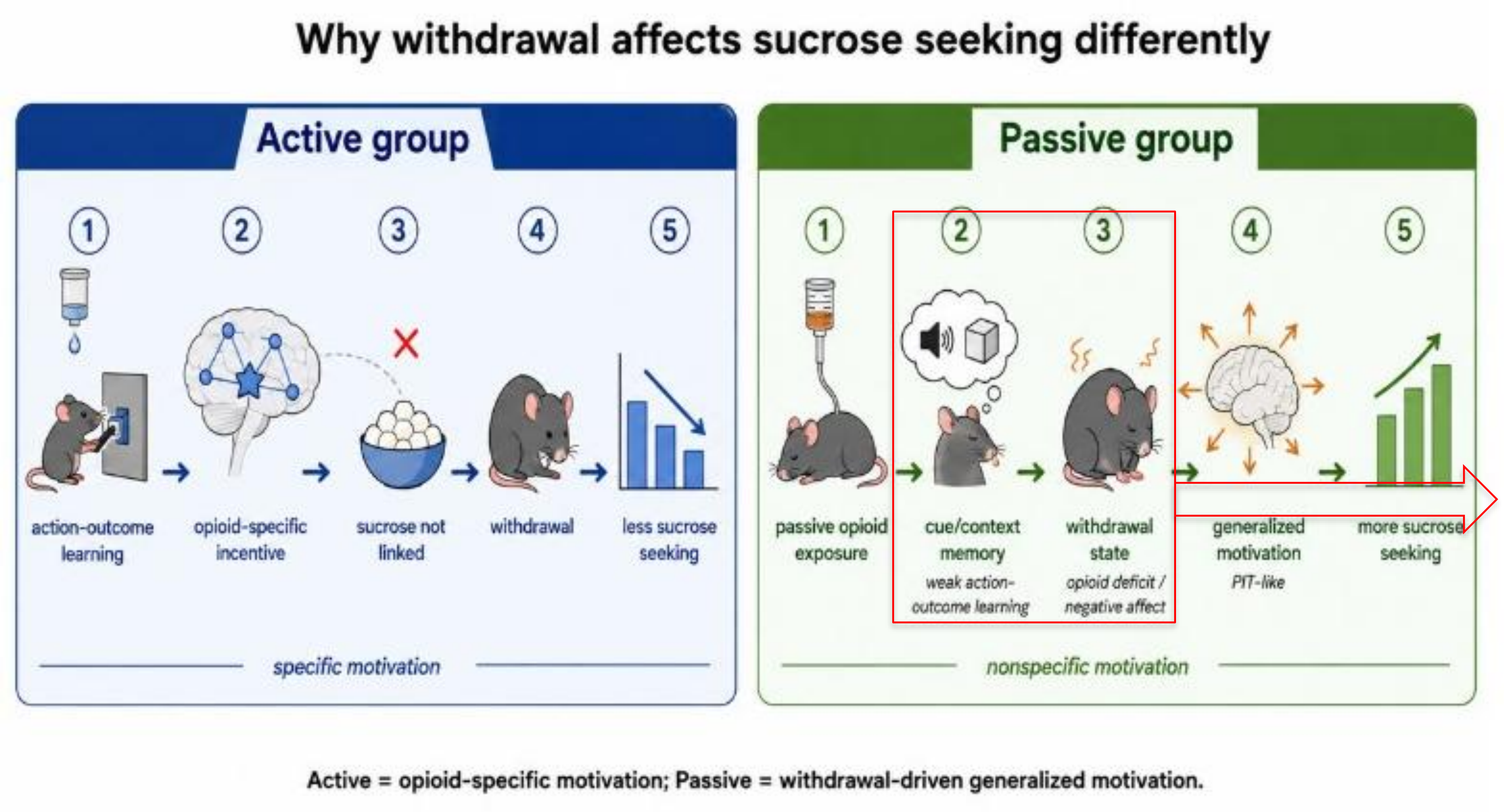
Whole-brain TRAP — another candidate stories

B) Passive withdrawal (patient-relevant)

second finding — why keep both

- FINDING B — COAa: Passive > Active at Post and Withdrawal (patient-relevant). Best anatomical fit in cluster 3 for cue/context + withdrawal-state → generalized motivation candidate (same numeric pattern also LGv, FC).
- COAa = best anatomical fit for Finding B schematic (cue/context + withdrawal-state → generalized motivation candidate).
- LGv (visual thalamus) & FC (hippocampal remnant) = same pattern, weaker anatomy for PIT / affect story.

Finding B — Why Passive matters (patient-relevant)



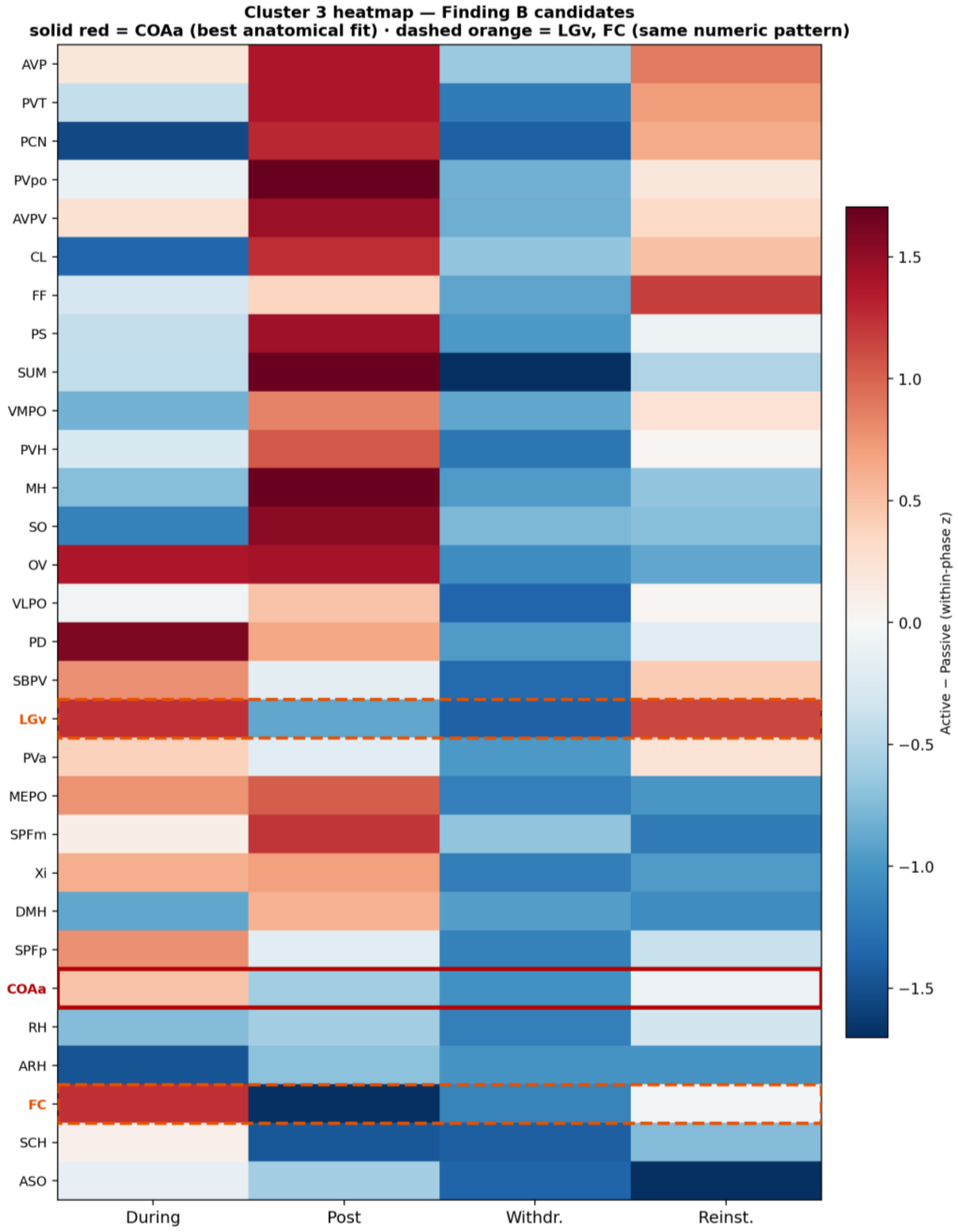
- Active: opioid-specific → less sucrose seeking in withdrawal.
- Passive: generalized motivation → MORE sucrose seeking.
- Behavior: withdrawal ↑ sucrose is Passive-only.

Candidate: cue/context (Post) + withdrawal-state → generalized (PIT-like) motivation.

Passive > Active at Post and Withdrawal in COAa (best anatomy in c3; also LGv/FC numerically).

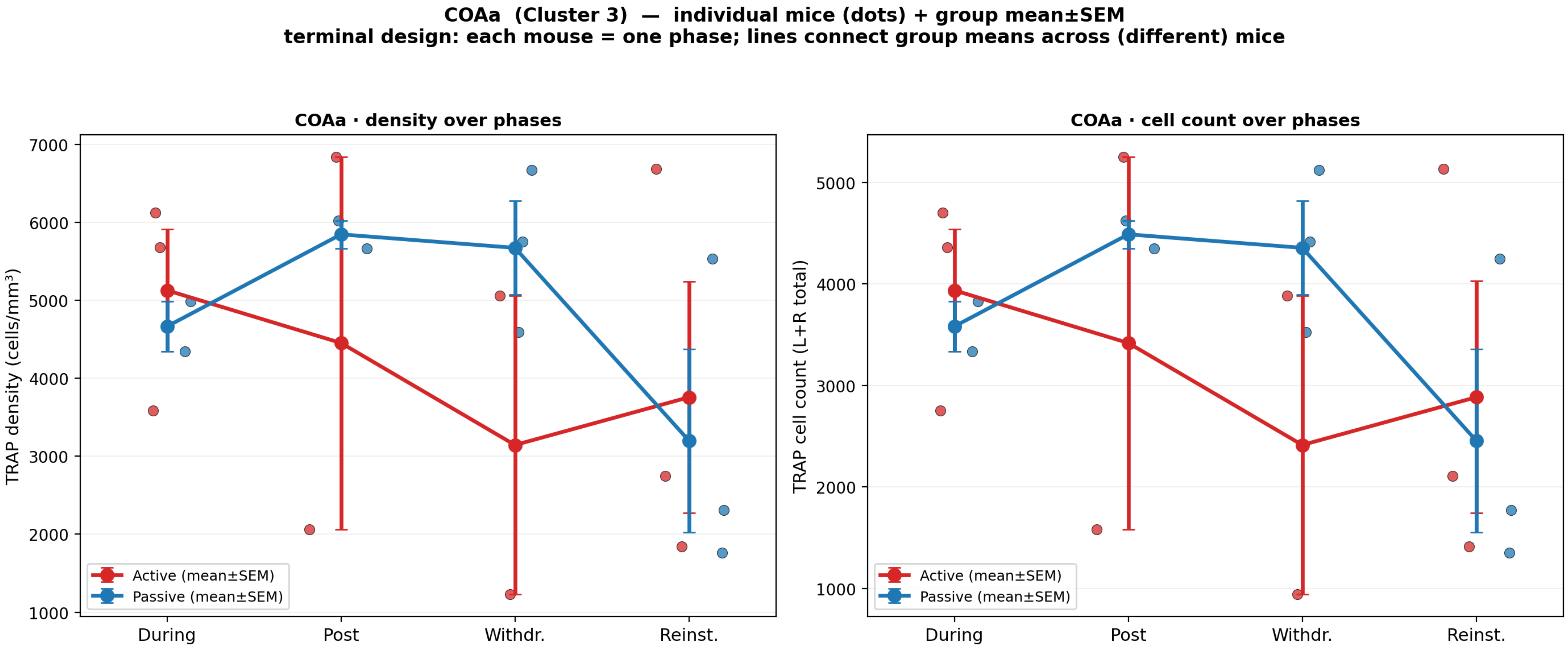
- Patients often first exposed noncontingently → Passive is translational.

Cluster 3: heatmap — Finding B candidates



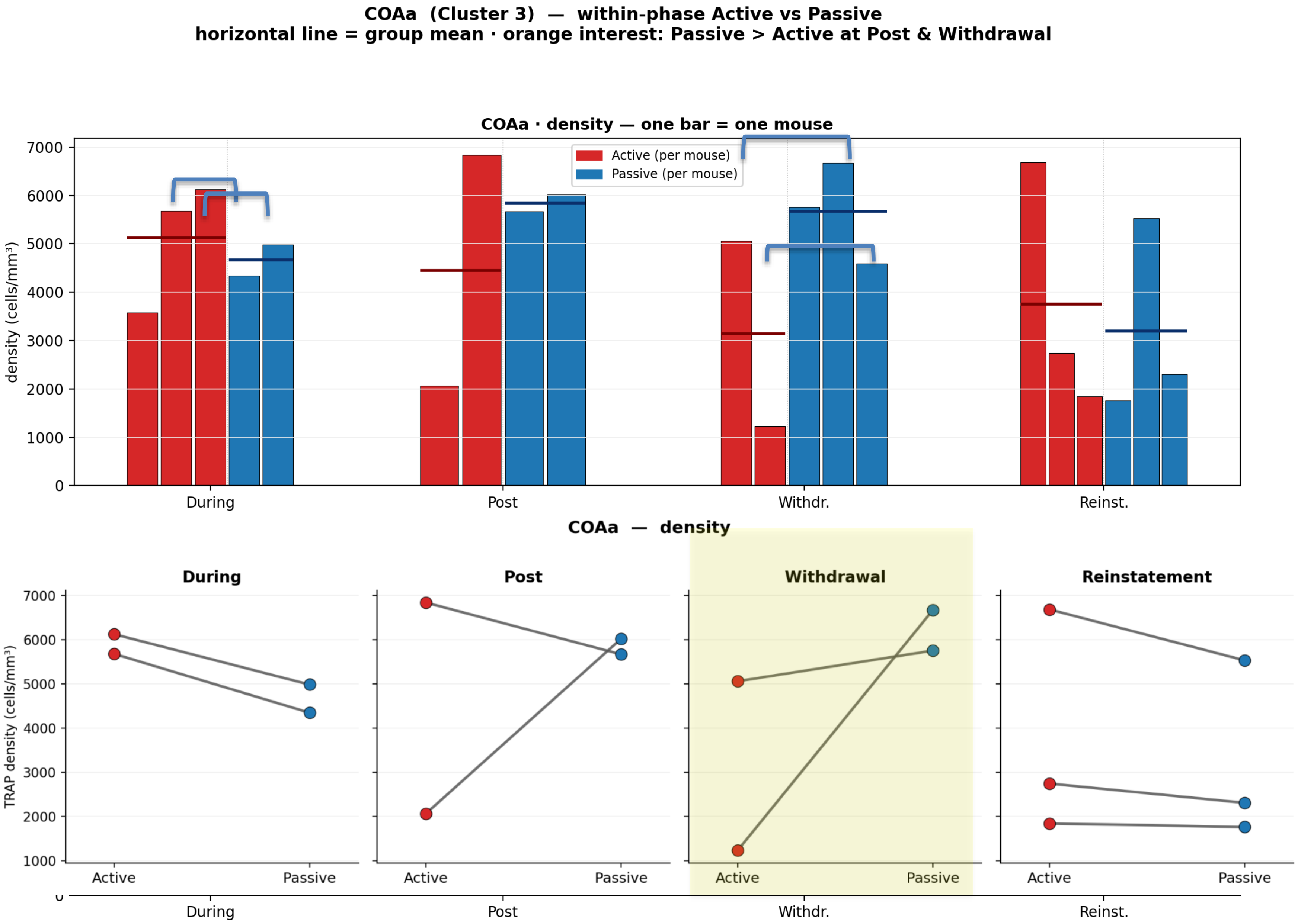
- Red box = COAa (chosen).
- Orange dashed = LGv, FC (numeric peers).
- Metric: Active − Passive (z).

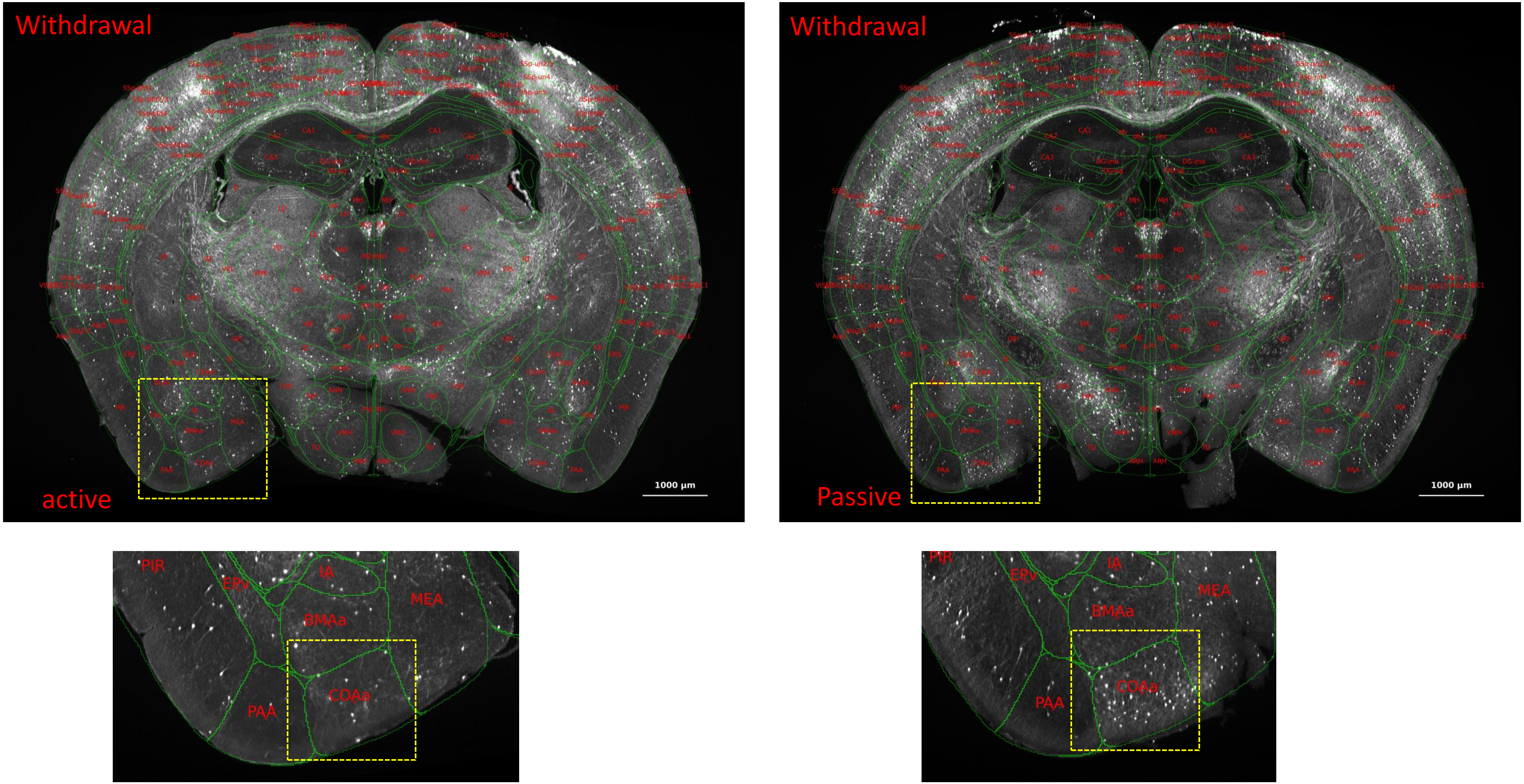
Finding B — COAa zoom-in (density & cell count)

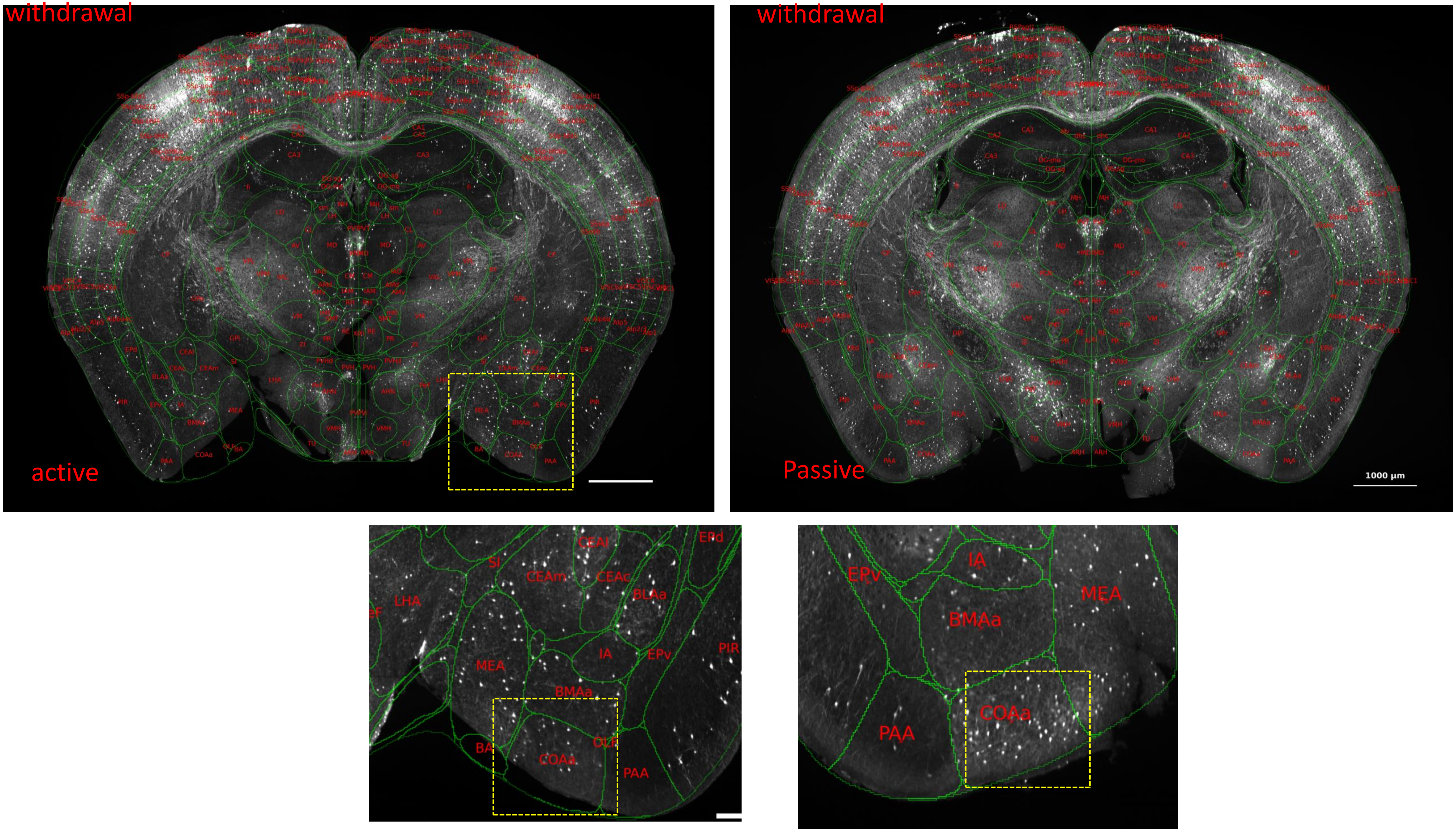


dots = individual mice (Active red / Passive blue) · no per-mouse cross-phase lines (TRAP is a terminal snapshot, not longitudinal)

Finding b — COAa within-phase (one bar = one mouse)




Withdrawal
active
1000 µm
Withdrawal
Passive
1000 µm
PIR
ERv
IA
MEA
BMAa
COAa
PAA
PIR
ERv
IA
MEA
BMAa
COAa
PAA


withdrawal
active
withdrawal
Passive
1000 μm
CEAl
EPd
SI
CEAm
CEAc
BLAa
LHA
MEA
IA
EPv
PIR
BMAa
BA
OLF
COAa
PAA
IA
EPv
MEA
BMAa
COAa
PAA

Summary

- A) ORBm & BMAp — dual-role craving: opioid seeking + negative-affect in Withdrawal (Active >> Passive).
- B) COAa — Passive > Active at Post and Withdrawal (patient-relevant); best anatomical fit in c3
- Keep both → complementary Xenium / chemogenetic targets (A: Active craving ORBm/BMAp; B: Passive Post+WD bias COAa).

Activating COAa should bias toward broad reward seeking; activating ORBm/BMAp should bias toward opioid-related seeking. That's the dissociation I predict — still needs a causal test.